



Marine Biology

海洋生物学



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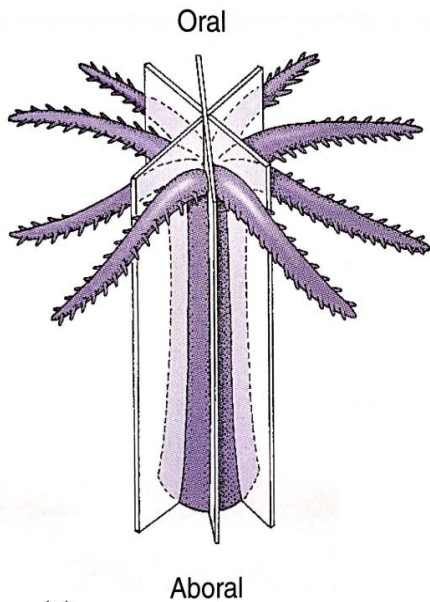
• Major Phyla of Marine Invertebrates

- | | |
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| 1. Sponges (海绵动物) | 10. Arthropods (节肢动物) |
| 2. Cnidarians (腔肠动物) | 11. Bryozoans (苔藓虫) |
| 3. Flatworms (扁形虫) | 12. Phoronids (帚虫) |
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| 9. Molluscs (软体动物) | 18. Lancelets (文昌鱼) |

3. The Cnidarians (sting cells animals, Cnid: nettle)

3.1 Characteristics of Phylum:

- Radial symmetry, allows them to sting and capture organisms in any direction
- Mostly marine organisms, about 10,000 species known
- Generically, includes Corals, Sea Anemones and Jellyfish



Elk Horn Coral

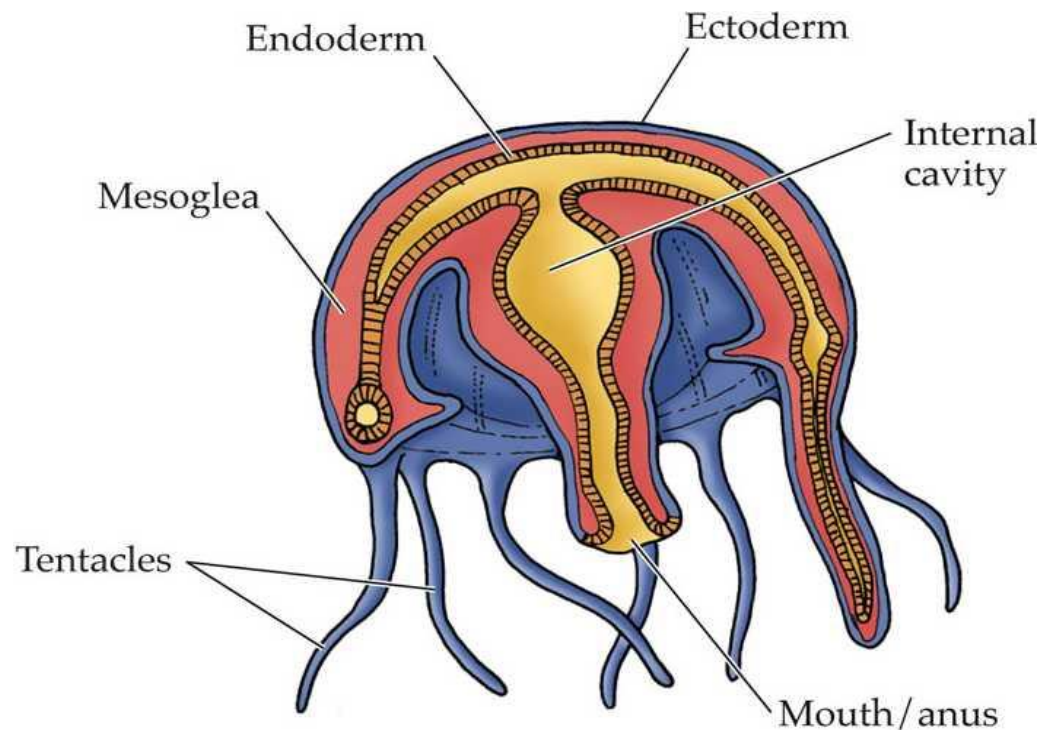


Sea Anemone



Jellyfish

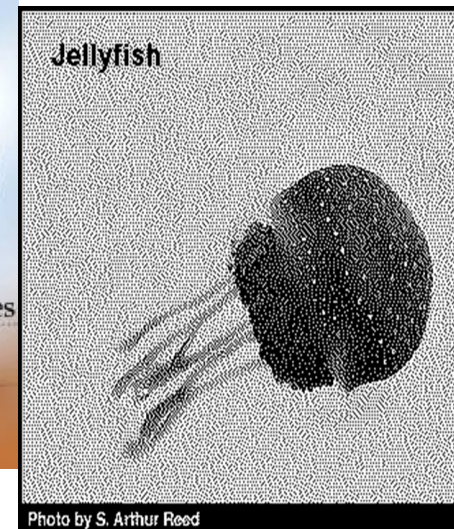
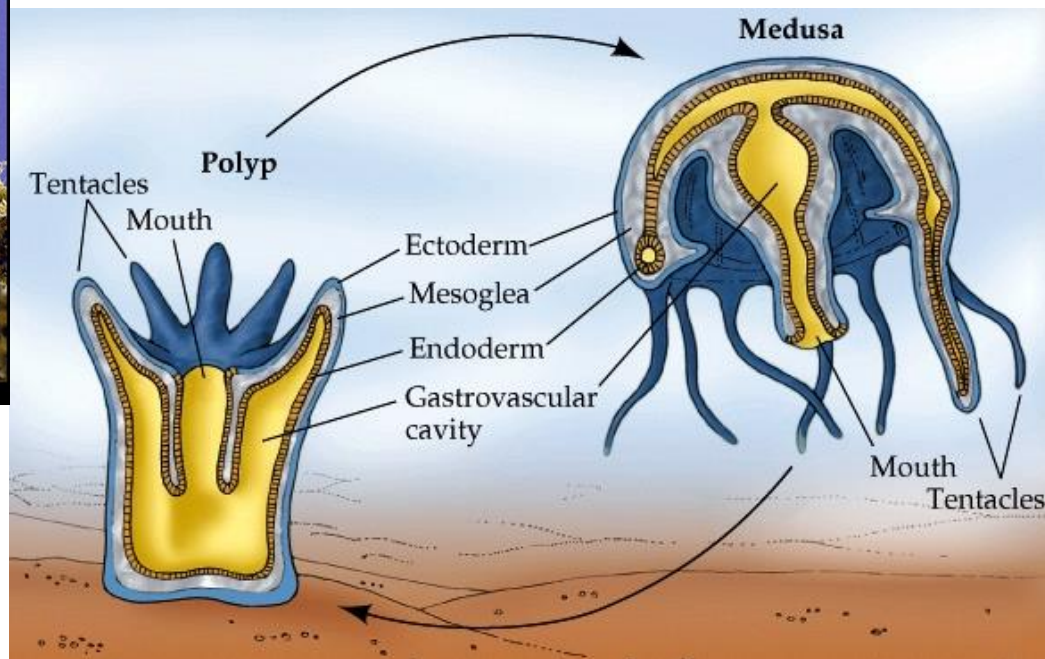
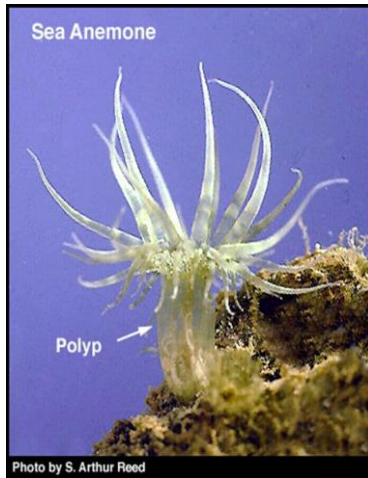
- Outer layer: **ectoderm/epiderm** (表皮) has cells that capture food and secrete mucus
- Jelly-like material between two layers: **Mesoglea** (中胶层)
- Inside layer: **endoderm** (胃皮、内皮) surround the central cavity
- Central cavity: **gastrovascular/internal cavity** (消化循环腔) which extend to **tentacles** (触角)



– Two body forms exist :

□ Medusa (水母体): free floating form which is transported by water currents, mouth with surrounding tentacles are positioned downward

□ Polyp (水螅体): sessile, attached form with mouth and tentacles are positioned upward

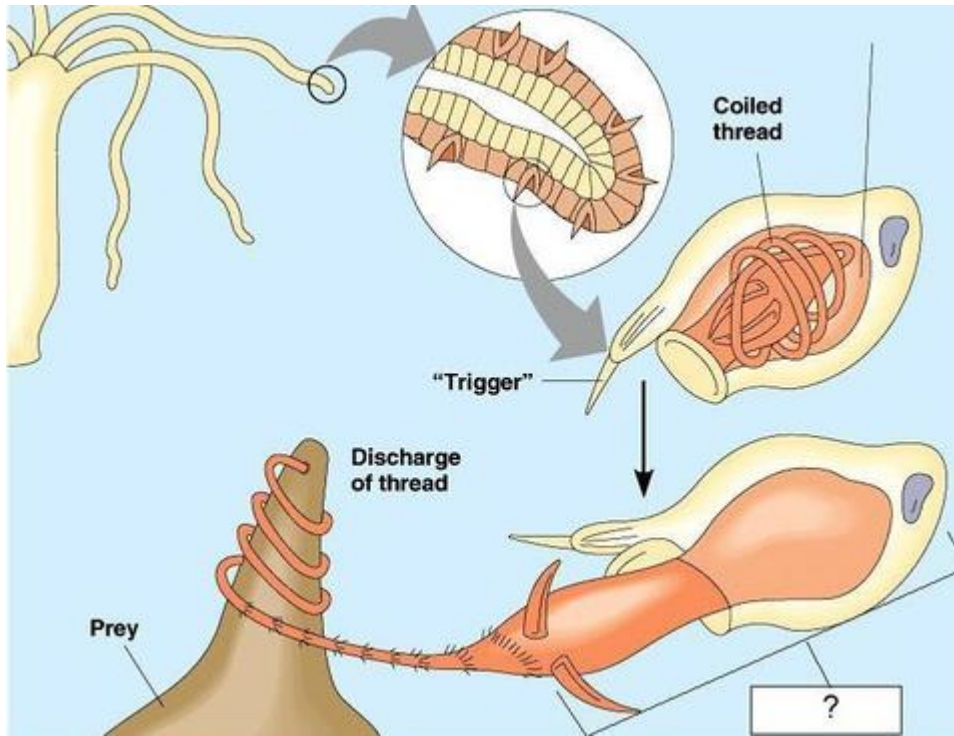


3.2 Food Capture

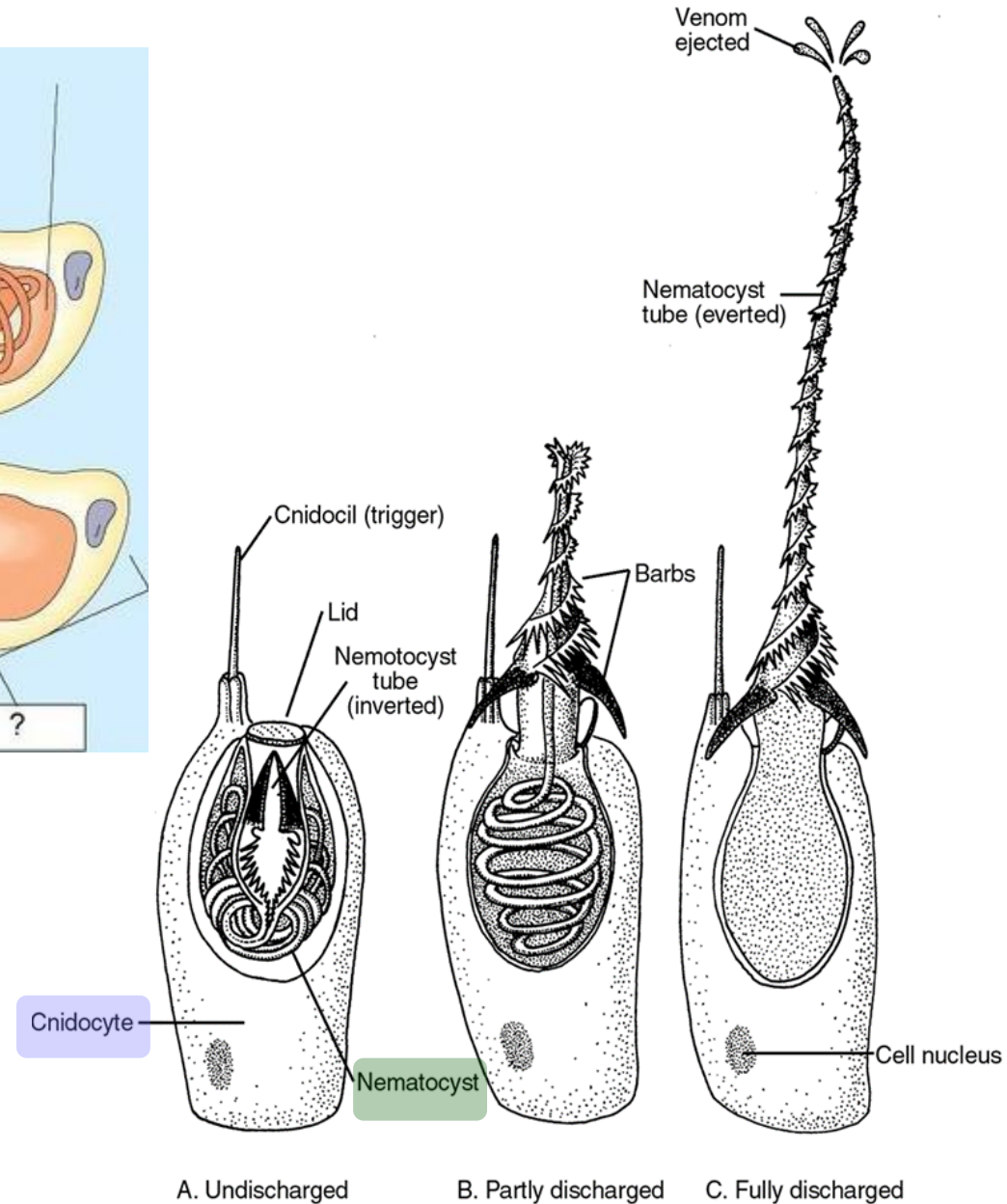
- Unique long stinging cells called: **Cnidocytes** (刺细胞)/nematocyte
- Long harpoon like tubes called: **Nematocyst** (刺丝囊)/cnidocyst

CNIDOCYTE VERSUS NEMATOCYST

Cnidocyte refers to a specialized cell, containing cnidocyst, which is involved in capturing the prey in cnidarians	Nematocyst refers to an organelle inside the cnidocyte, consisting of an ejectable thread that causes a sting
An ectodermal cell in cnidarians	Organelle that occurs inside the cnidocyte
Also called cnidoblasts or nematocytes	Also called cnidocyst or cnida
Stinging cells used in defense and capturing prey	Contains toxins to be injected into the predator or prey

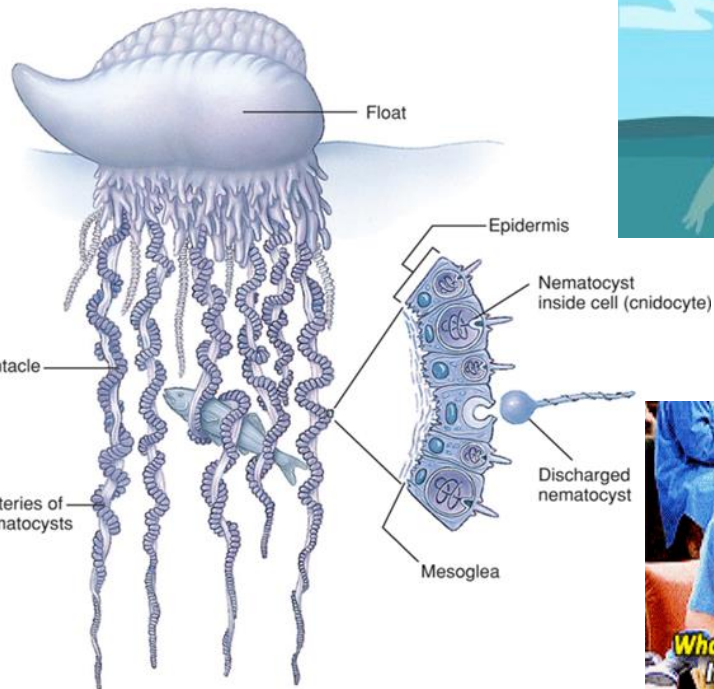


- When the Nematocyst senses food, it fires venom outward to stun the prey
- Barbs (倒钩) line the harpoon-tube to attach prey

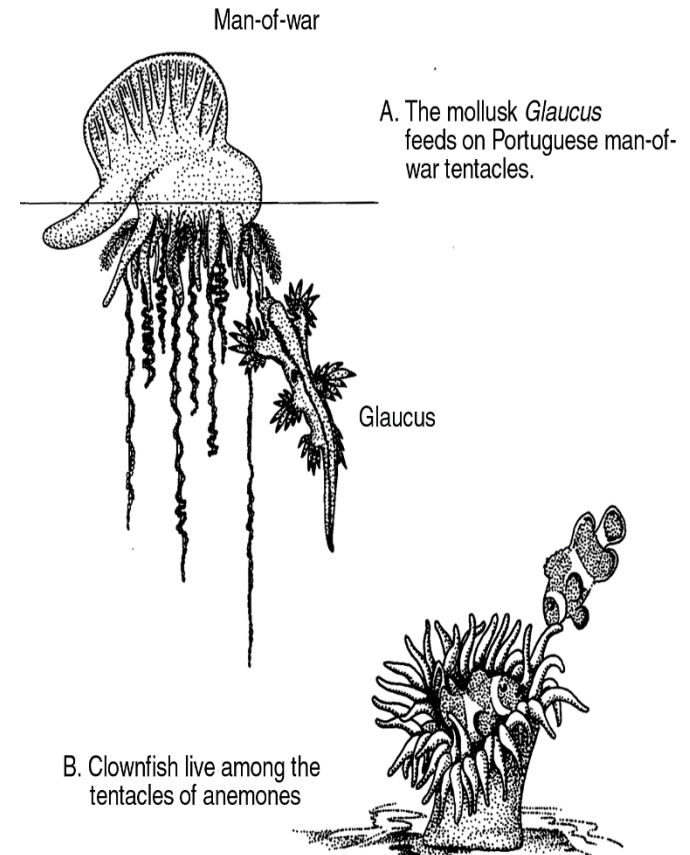


3.3 Stings

- Jellyfish sting to paralyze and capture food
- Venom is a protein, some people are allergic thus having a worse reaction
- Treat a sting by breaking down the protein

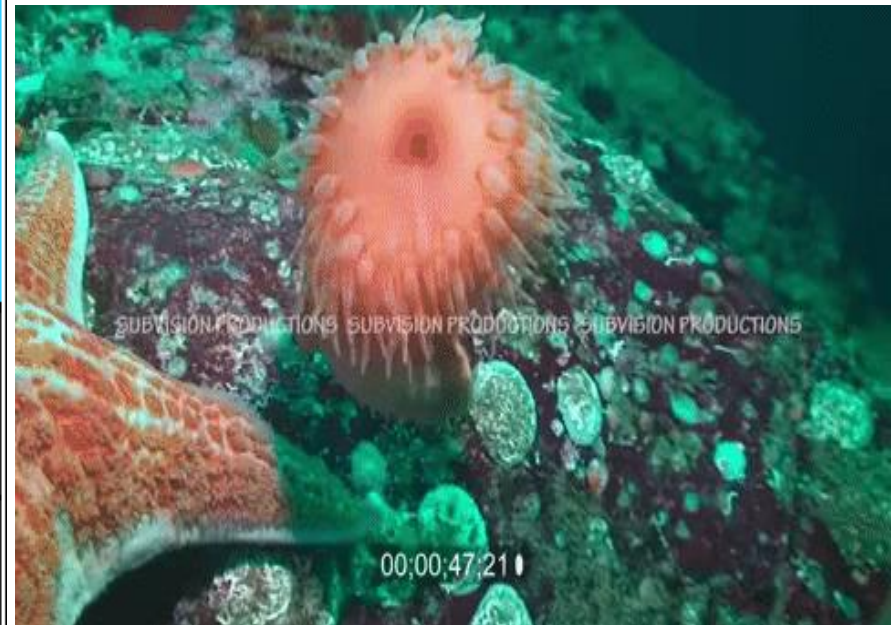
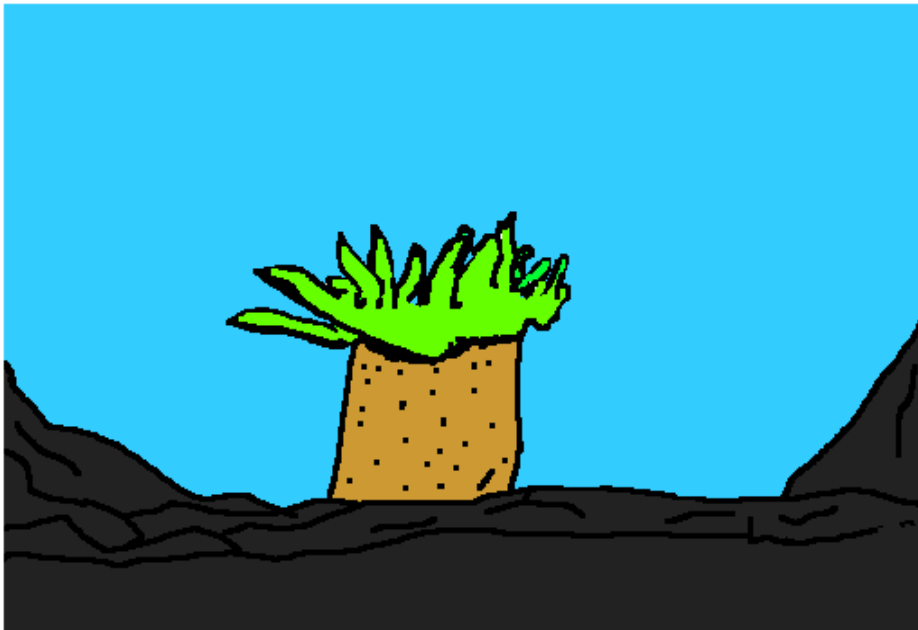


- Some marine organisms are immune, i.e. Clownfish, Glaucus (海神海蛞蝓), Turtles



3.4 Movement and Nervous System/Muscular System

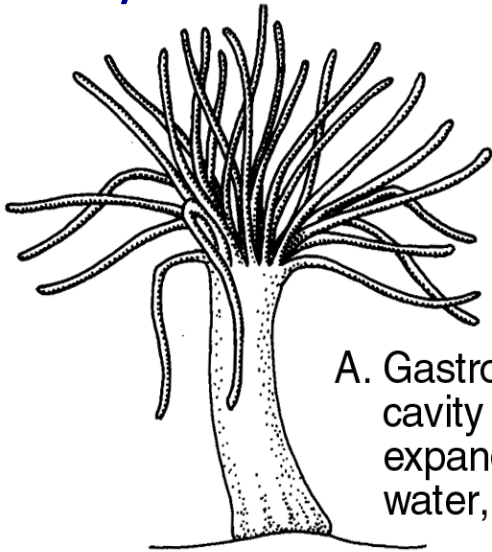
- sea anemones and jellyfish have **no skeletal structure** to support their soft tissue
- They fill gastrovascular cavity with water and close mouth tight
 - This puts the water under pressure like a balloon
 - If the Cnidarian contracts its body hard , it loses water. It takes several minutes to pump the water back in



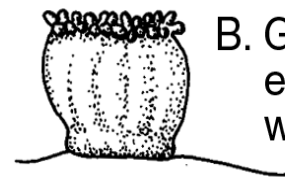
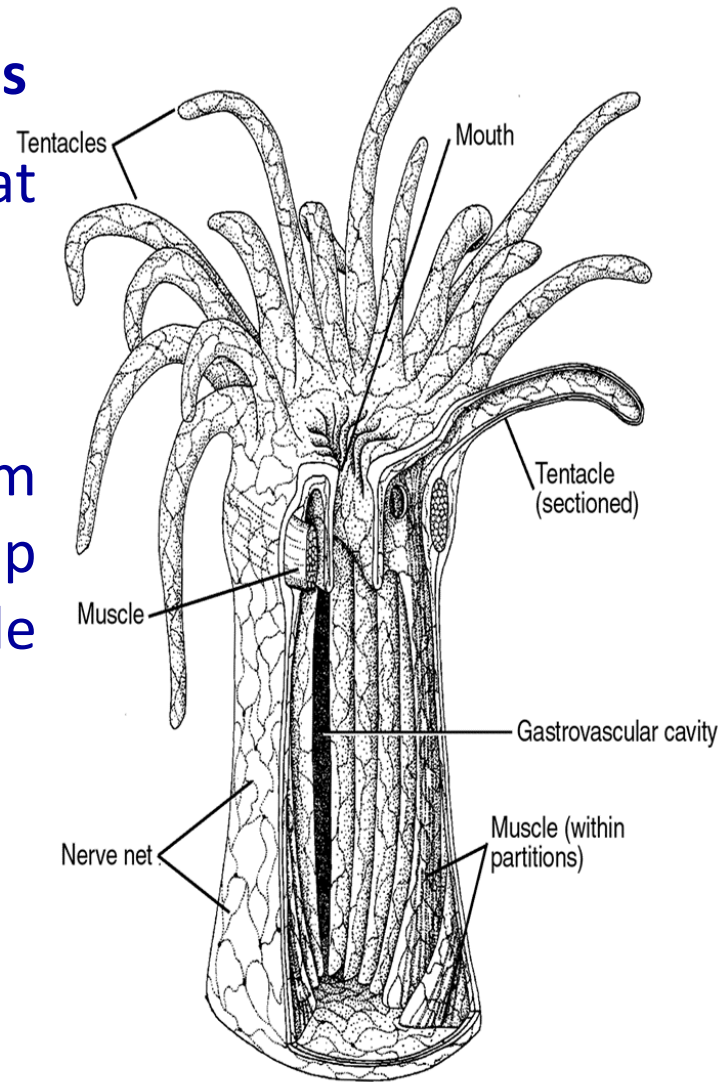
- A ring of muscles contract, a jet of water is forced out
- The jellyfish relaxes and then the umbrella reforms
- Alternating muscles contract creates pulsating movements
- Jellyfish are poor swimmers and are normally classified as plankton



- The movement is regulated by **nerve cells**
- Consists of cells with long, thin fibers that respond to stimuli
- Nerves send impulses to muscle cells
- In the sea anemone, muscular system consists of muscle cells lying in bands up and down the body wall and in a circle around the mouth
- Body shortens when contractions occur



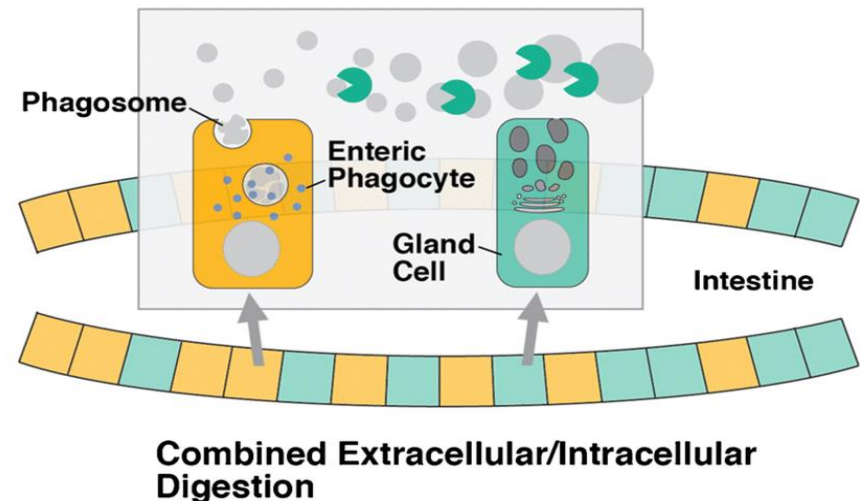
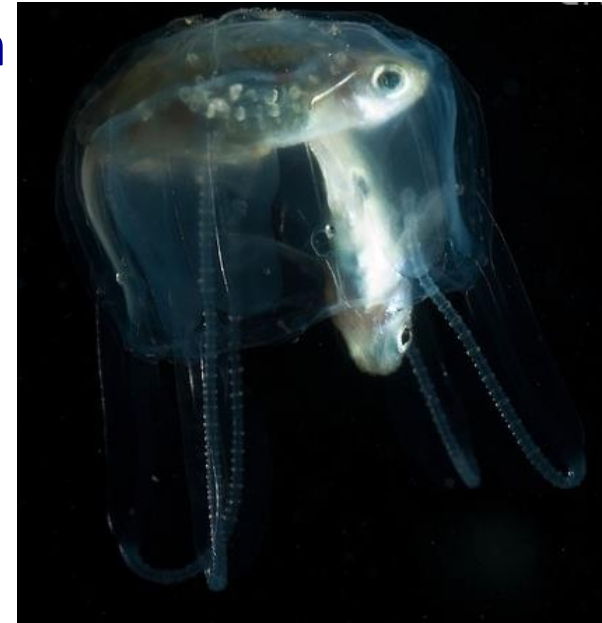
A. Gastrovascular cavity and tentacles expanded, filled with water, and relaxed



B. Gastrovascular cavity empty, tentacles and body wall contracted

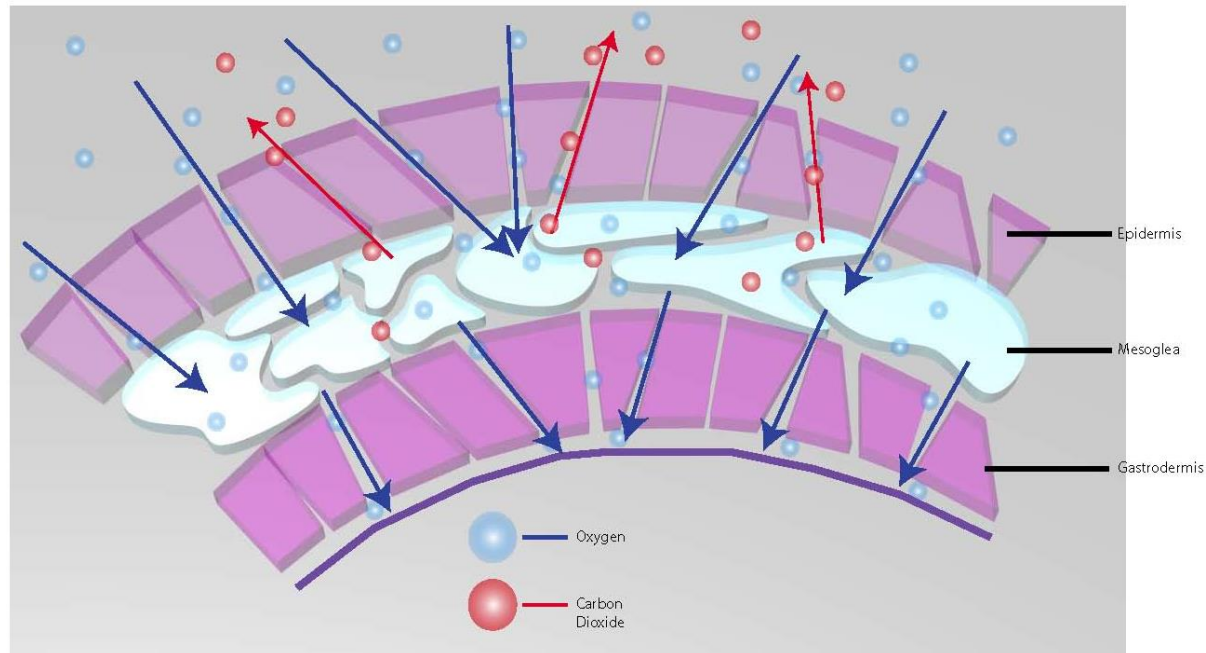
3.5 Digestion

- Digestive system is **incomplete** (sac-like with mouth only)
- Tentacles move the food into their mouths
- Prey is swallowed into the gastrovascular cavity and digested (extracellular digestion)
- Endoderm cells produce enzymes, which break down the food (intracellular digestion)
- Can consume anything from plankton to small fish
- Indigestible material is expelled through the mouth



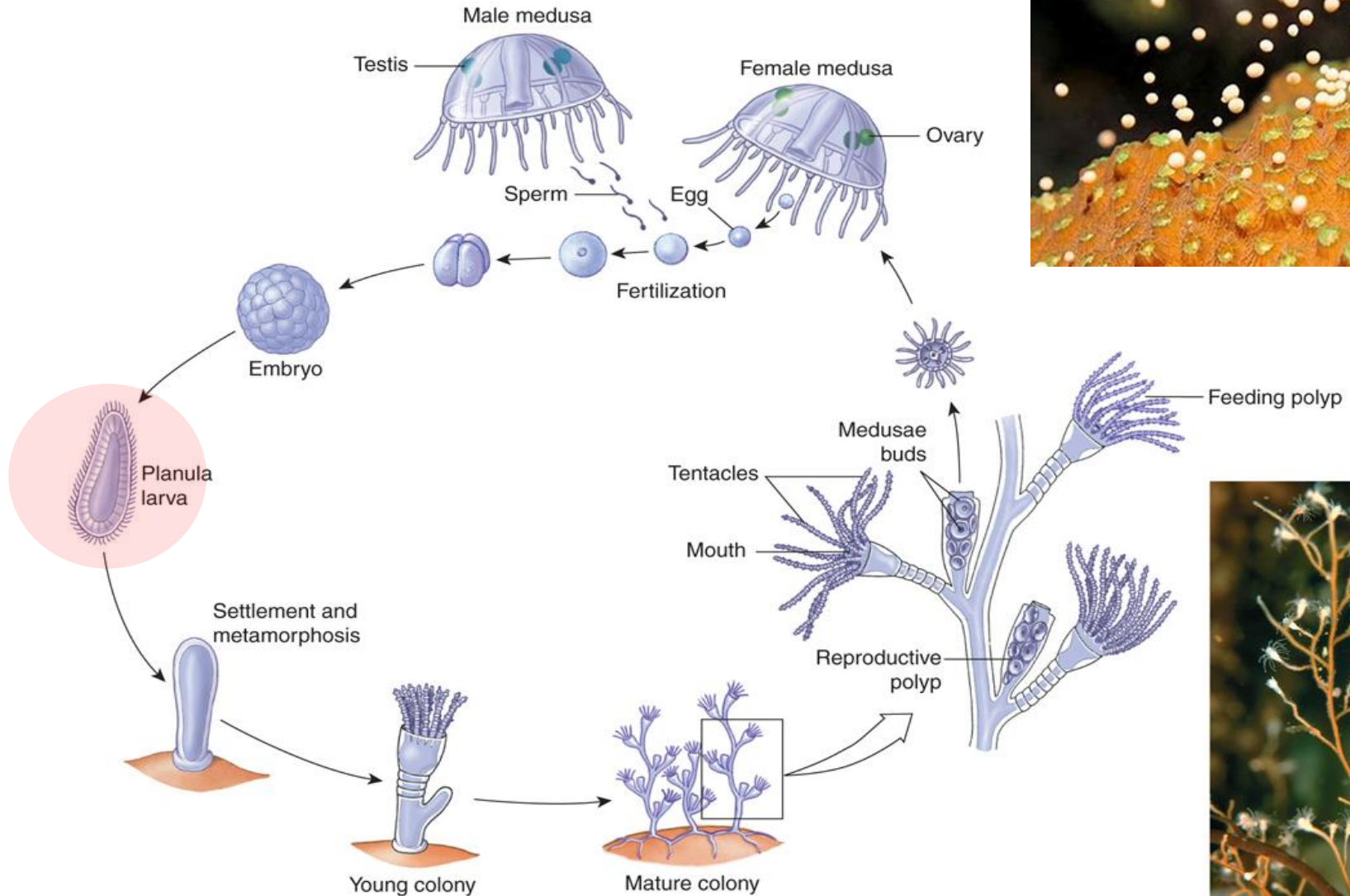
3.6 Respiration and nutrients absorption

- Have **neither respiratory or circulatory systems**
- Get oxygen directly from water
- Nutrients pass through the liquid between the cells, wastes pass out by the same route



3.7 Reproduction & Regeneration in Cnidarians:

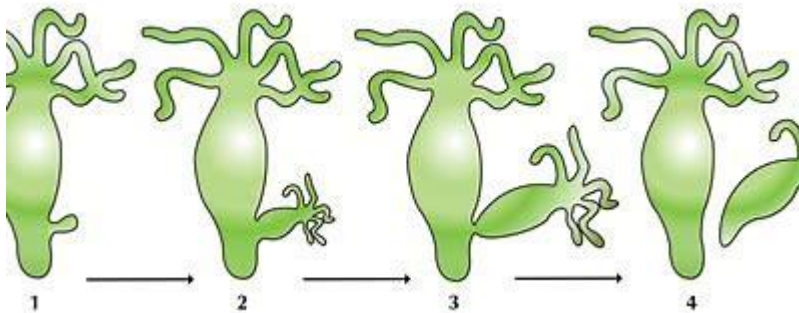
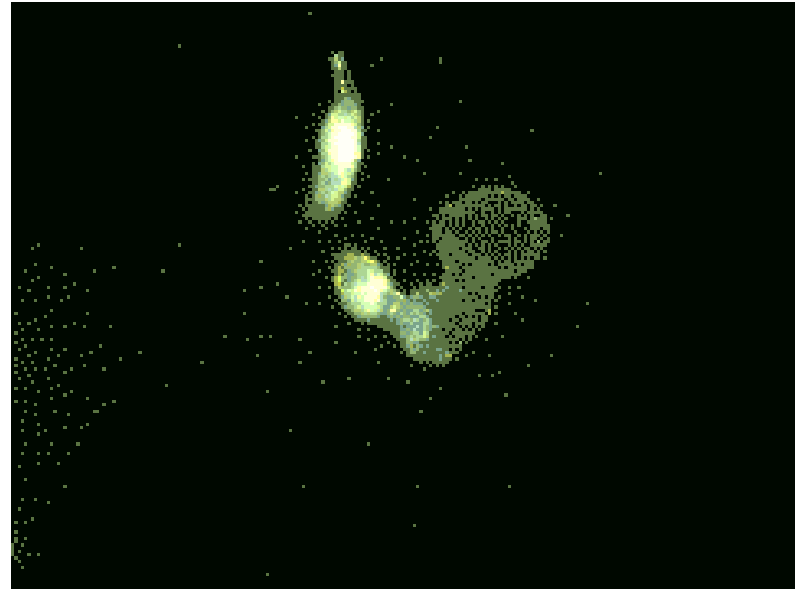
- Sexual reproduction



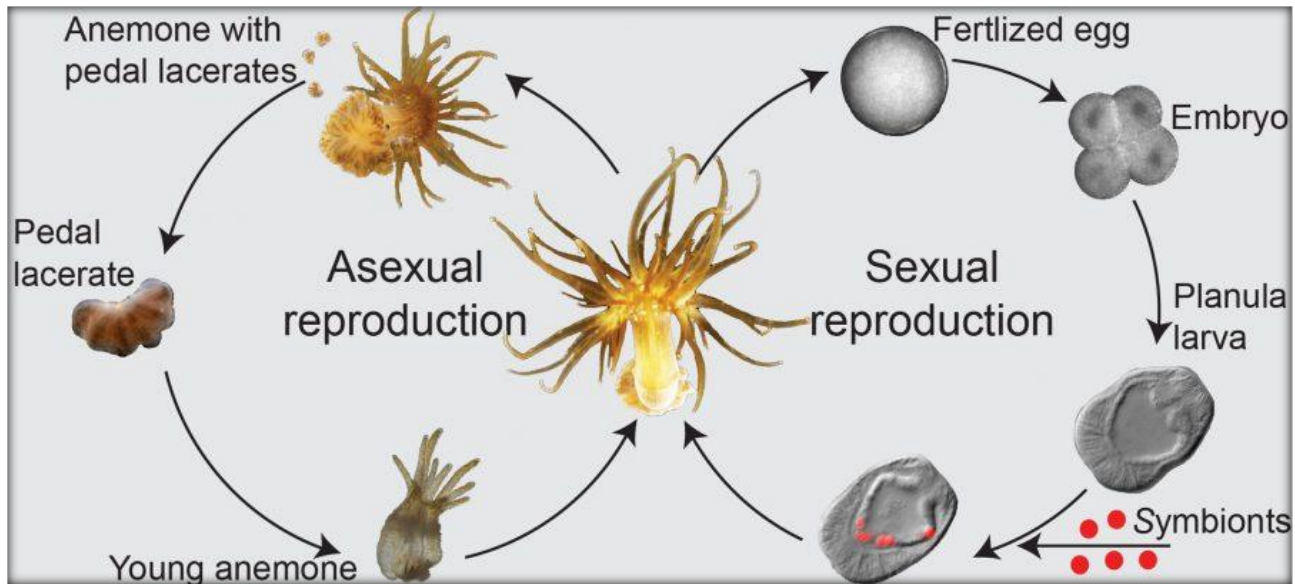
Life Cycle of a hydrozoan

Eudendrium, worldwide

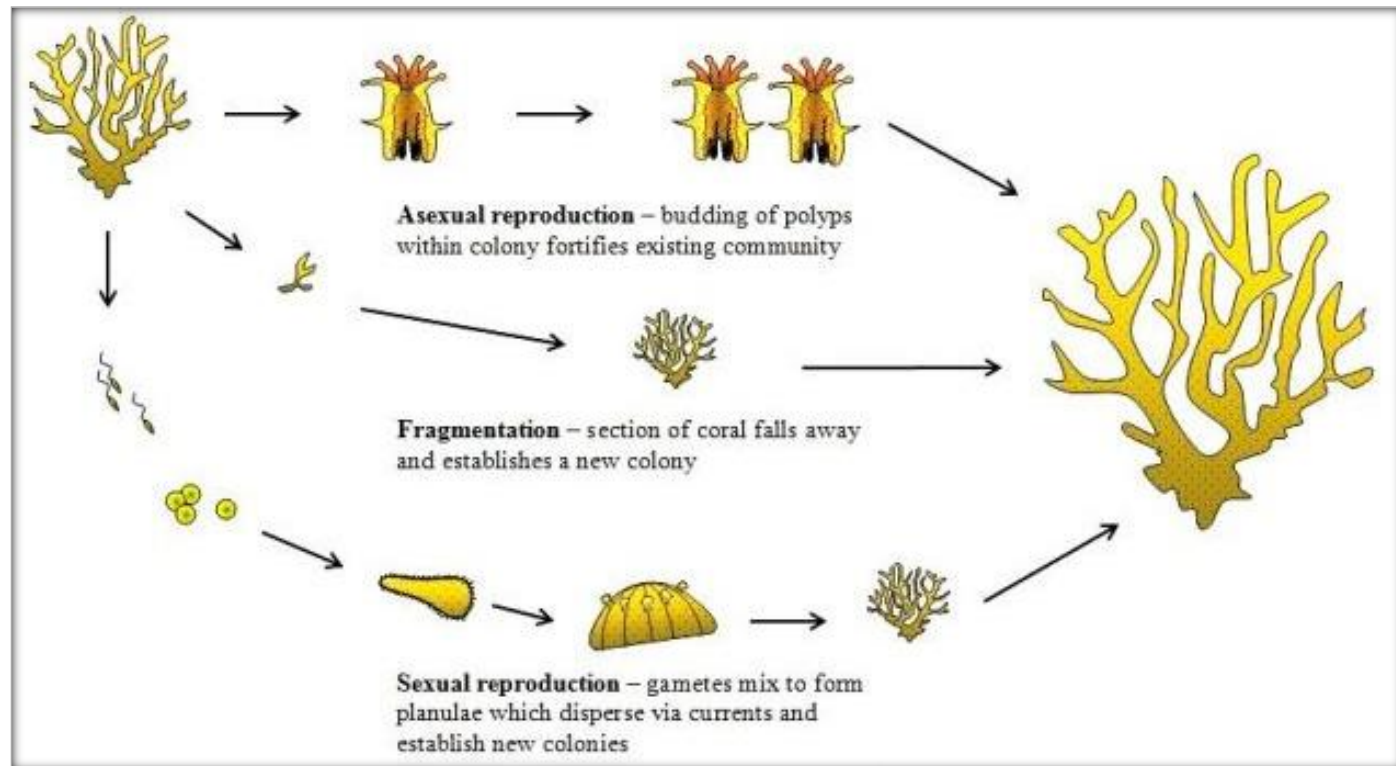
- Asexual reproduction: Polyps normally reproduce by budding
- Regeneration: Replace lost/damaged part; small damaged chunks may even regenerate into new organisms



Anemone



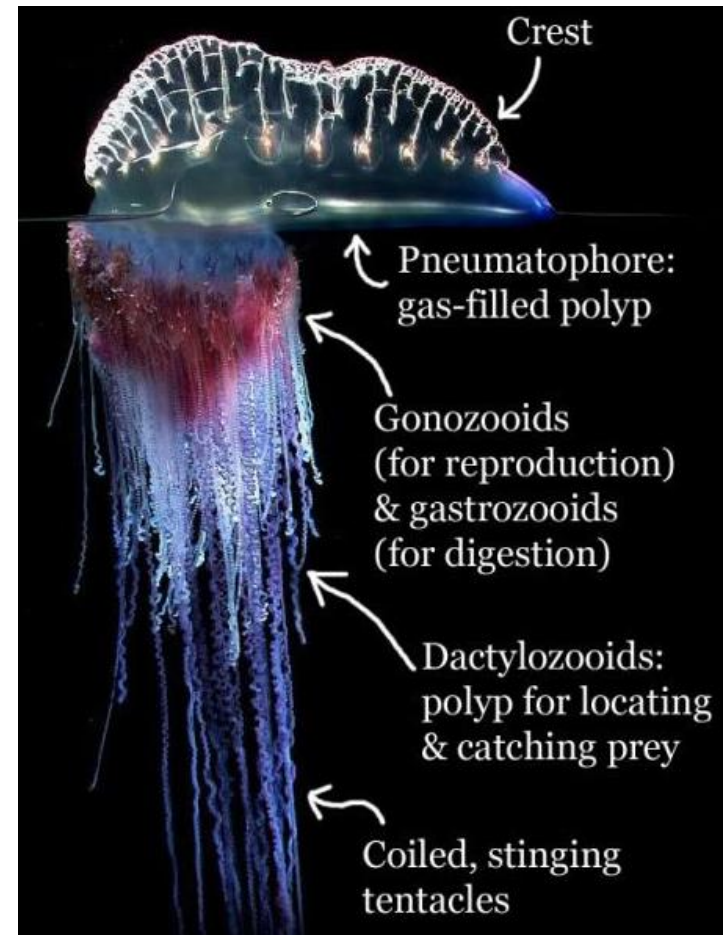
Coral reef



3.8 Classification of Cnidarians

1) Class Hydrozoa (水螅纲)

- Mostly polyp forms with reproductive medusa
- Physalia (僧帽水母) is unusual species with a colony of polyps carried by a gas-filled float
- Freshwater and marine species



2) Class Scyphozoa - True Jellyfish

- Free swimming large medusa forms with polyp **only** in reproductive life
- **All marine**
- They move by rhythmic contractions, but cannot fight against prevailing water currents, except for the one with very bigger size
- Many with powerful stings



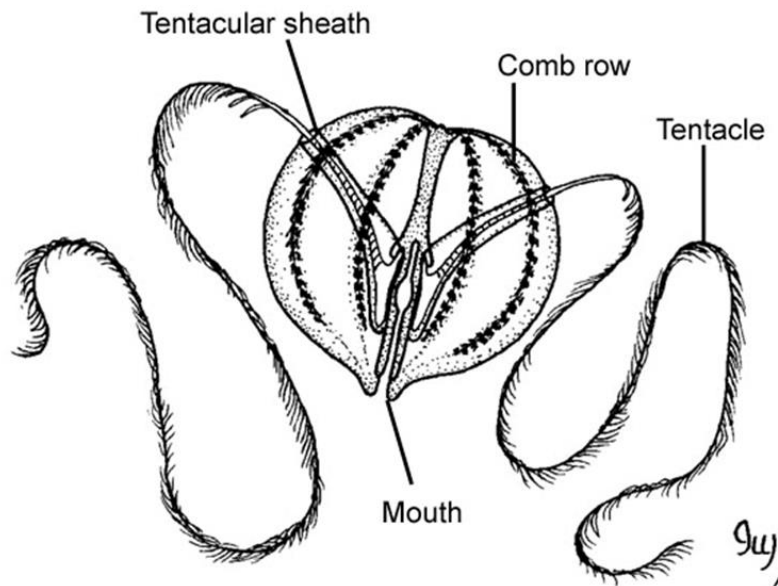
3) Class Anthozoa (珊瑚纲)– Corals, Anemones, Sea Fans/Whips, Sea Pansy

- Colonial polyps that normally lack a medusa stage
- Corals secrete calcium carbonate “shells”
- Most coral species possess symbiotic algae within body tissues called zooxanthellae (虫黄藻)
- **All marine**



3.9 Ctenophores (栉水母) – also known as comb jellies

- About 100 species, **all marine**, found in warm and cold waters
- **8 rows of thick cilia that beat continuously**
- No nematocysts unless they harbor some from eating cnidarians
- Long sticky tentacles used to capture prey
- Most species are planktonic

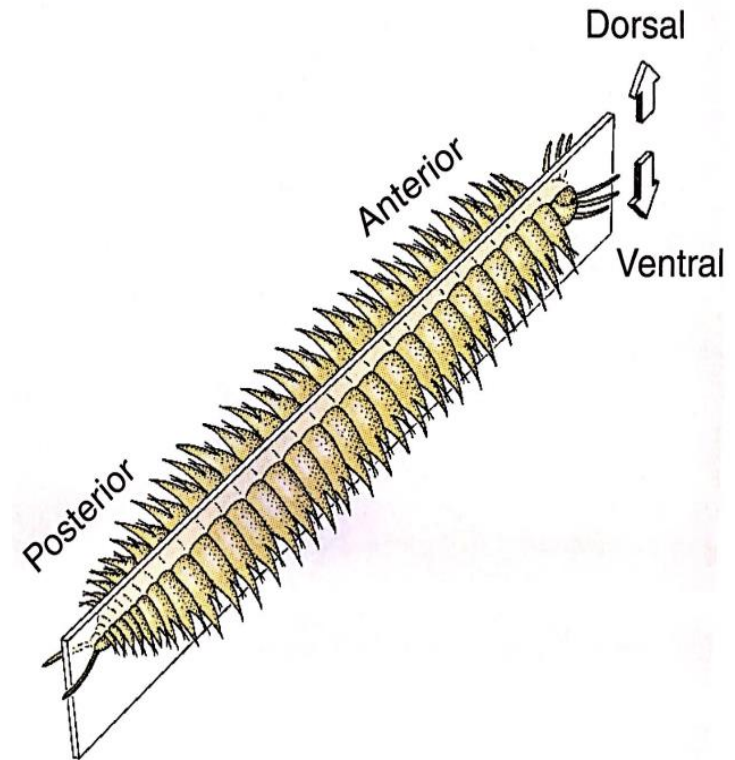
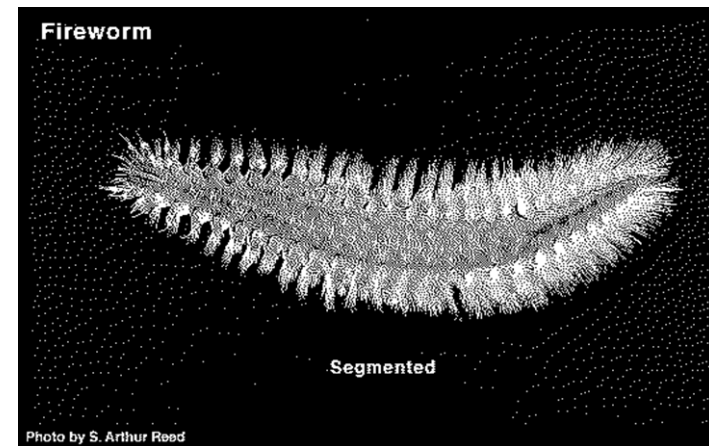


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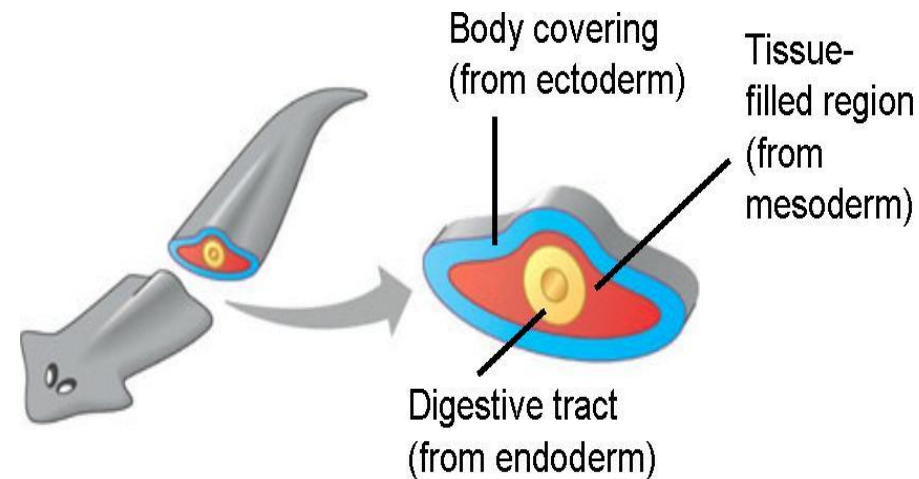
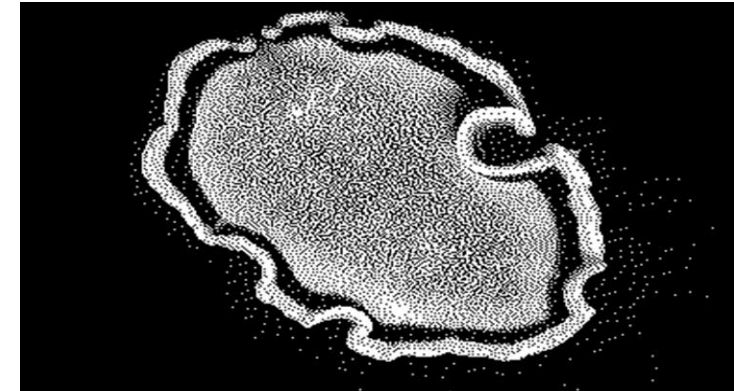
Invertebrate—The Worms

- Display a great range in size, efficiency, complexity, but share **long thin body structure** that get around efficiently without legs
- Different phyla = different degrees of complexity, which depends on different body features and systems
- Have **Bilateral symmetry**: front (the anterior end) and rear (the posterior end); dorsal surface and ventral surface
- At the anterior end is a head with a brain (at least an accumulation of nerve cells), and sensory organs, i.e. eyes



4. The Flatworms (phylum of Platyhelminthes, Simple Worms, 扁形虫)

- Least complicated, more complex than Cnidarians (腔肠动物, 刺细胞生物)
- Live on/in land, water, other animals
 - Symbiotic parasites (ex. Tapeworms)
 - Can make humans or animals sick or kill them
- Flatworms have 3 layers: ectoderm(表皮), endoderm(内皮/胃皮), and mesoderm(中胚层); Cells specialize into a **muscular (肌肉的) system** that enables the animal to move around



- **Have a simple CNS**

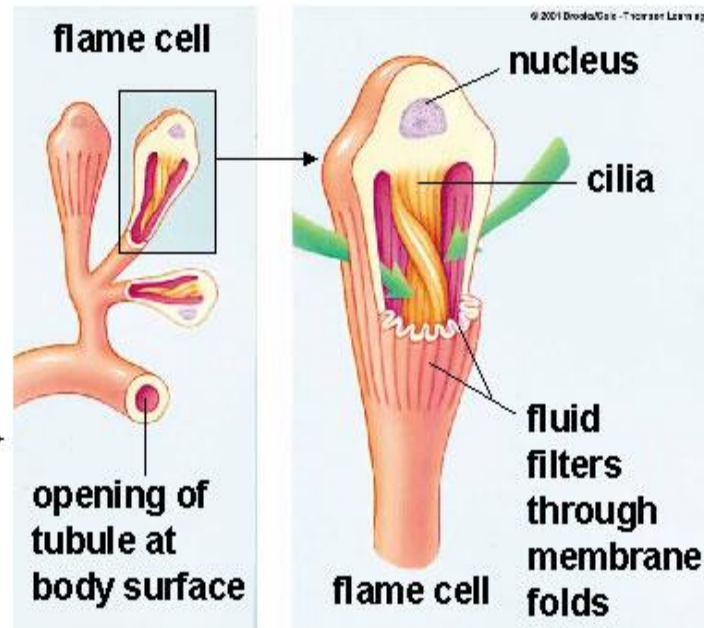
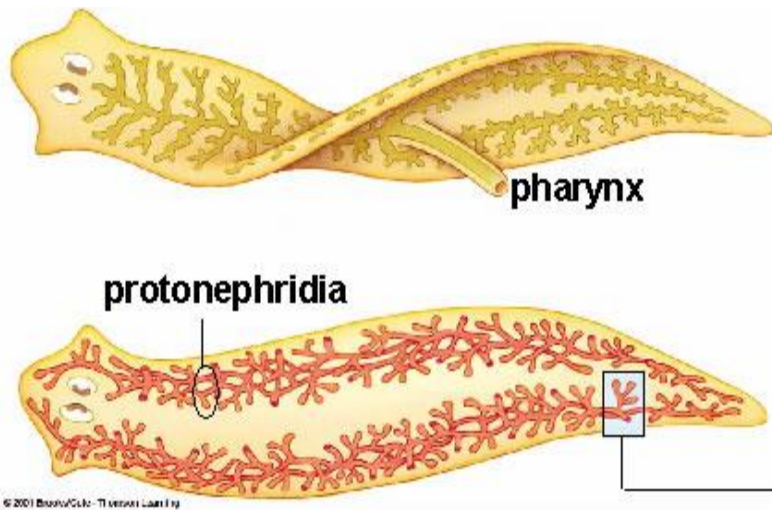
- Mass of nerve (sensory) cells called a brain and simple nerve cord along the body
- Sensory cells in the head detect changes in the environment:
Sensory cells that detect light are clustered into two eyes in the head;
Sensory cells that detect water currents, solid objects, and chemicals are in two flap like projections on the head called **auricles** (耳廓)

- **Have no circulatory system:**

- Get nutrients by diffusion
- Body shape and constant contact with water facilitates absorption
- All their cells are within (凋谢) close to their outer surface of digestive cavity

- Have **excretory system (protonephridia)**:

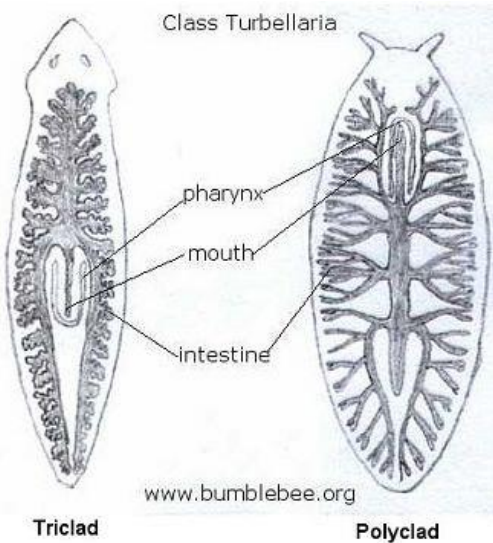
- Network runs the length of the animal and opens to the outside through small pores (opening of tubule) in the posterior region of the body
- Because this movement looks like a flickering fire, the structure is called a **flame bulb**



• Types of Marine Flatworms

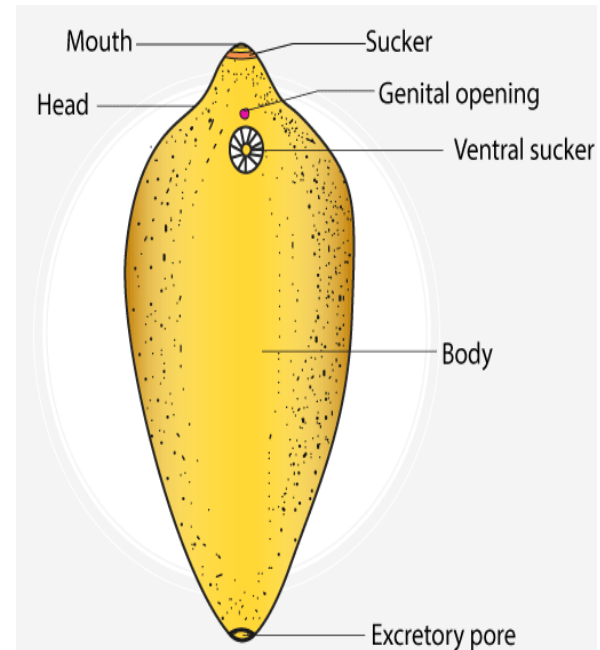
1) Turbellarians (涡虫)

- Mainly free-living carnivores
- Eye-spots present to determine light/dark patterns
- Most small
- Some with striking coloration patterns



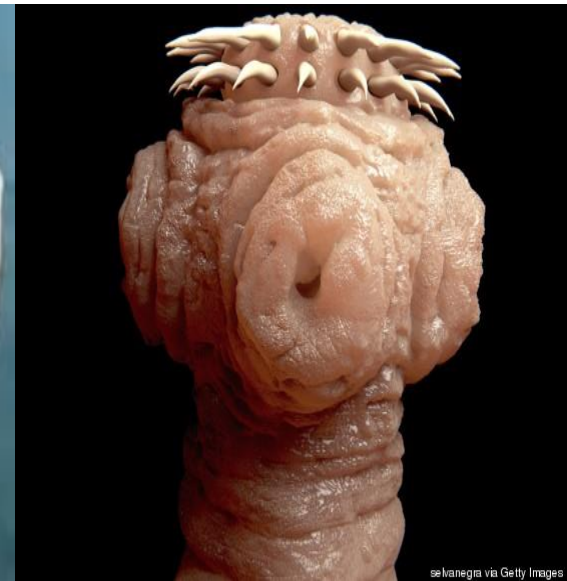
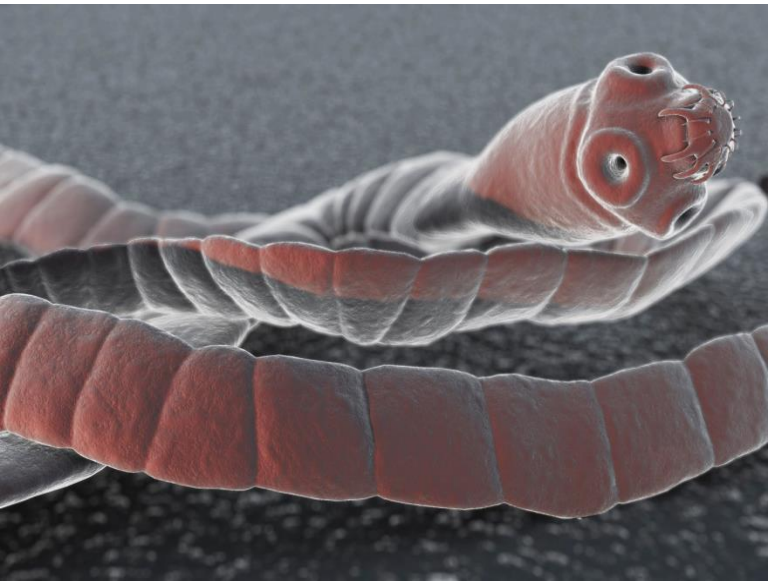
2) Flukes (吸虫)

- All parasitic
- Amazing reproductive abilities
- Vertebrates are the host of the adult fluke, have suckers to attach to inside of blood vessels near intestines of host or other tissues
- Larval stages may be harbored in shellfish or fish



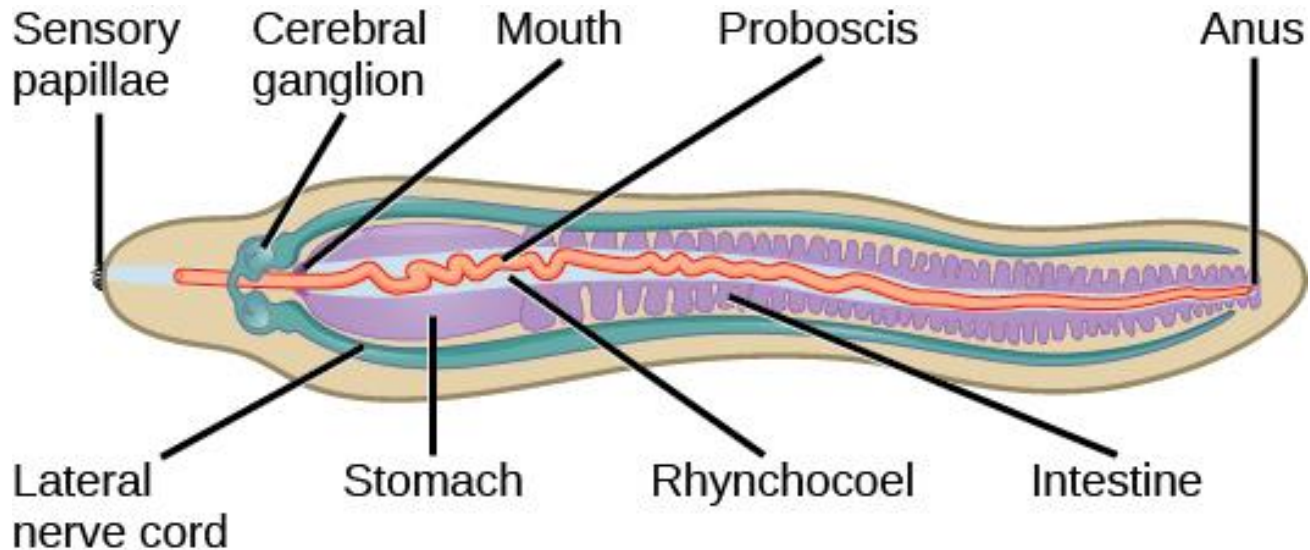
3) Tapeworms (绦虫)

- ❑ Parasitic, scolex (头节) with suckers and hooks for attachment in host's intestines
- ❑ Body is made of repeated segments
- ❑ Specialized cuticle surrounds body - allows absorption of nutrients
- ❑ No digestive system of their own – they get all their nutrients from their host

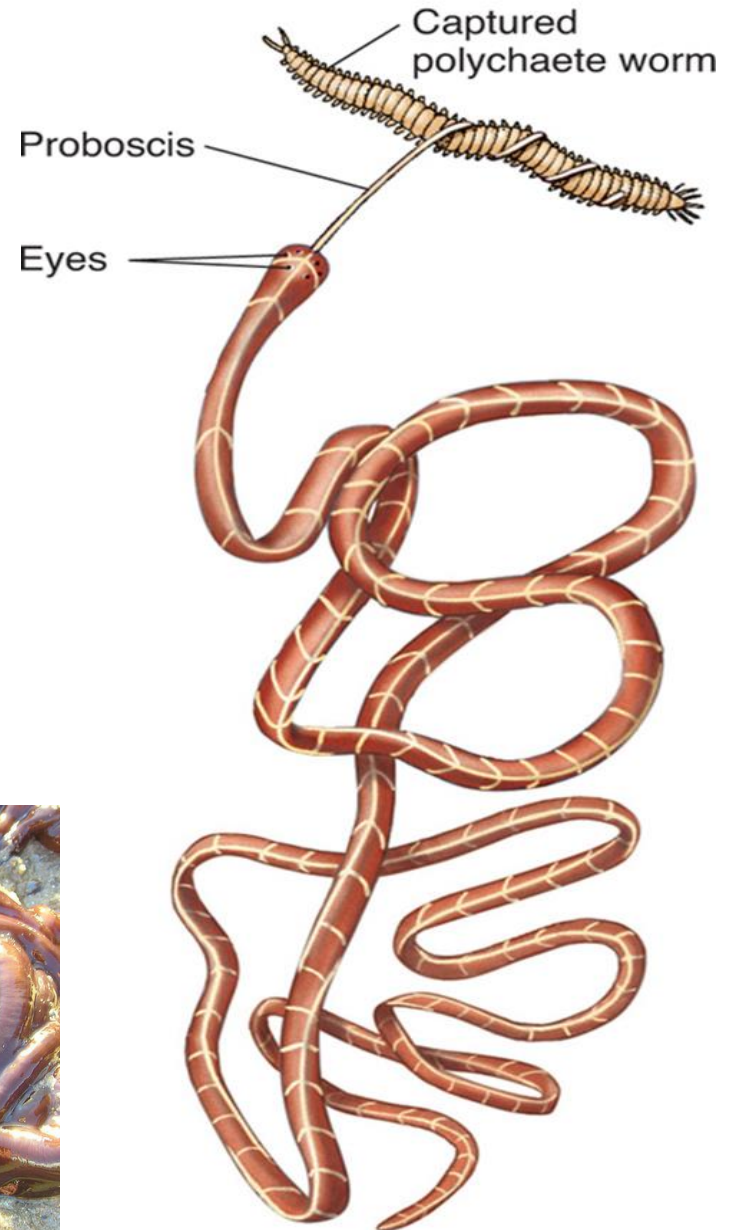


5. Ribbon Worms (纽虫, phylum of Nemertea)

- Mostly marine, prefer shallow, coastal waters
- Ciliated epidermis (表皮), proboscis(鼻), a long fleshy tube, is used to entangle prey
- **Complete digestive system with mouth and anus (肛门)**
- **Complete circulatory system:** the simplest animals to have a closed loop blood-vascular system

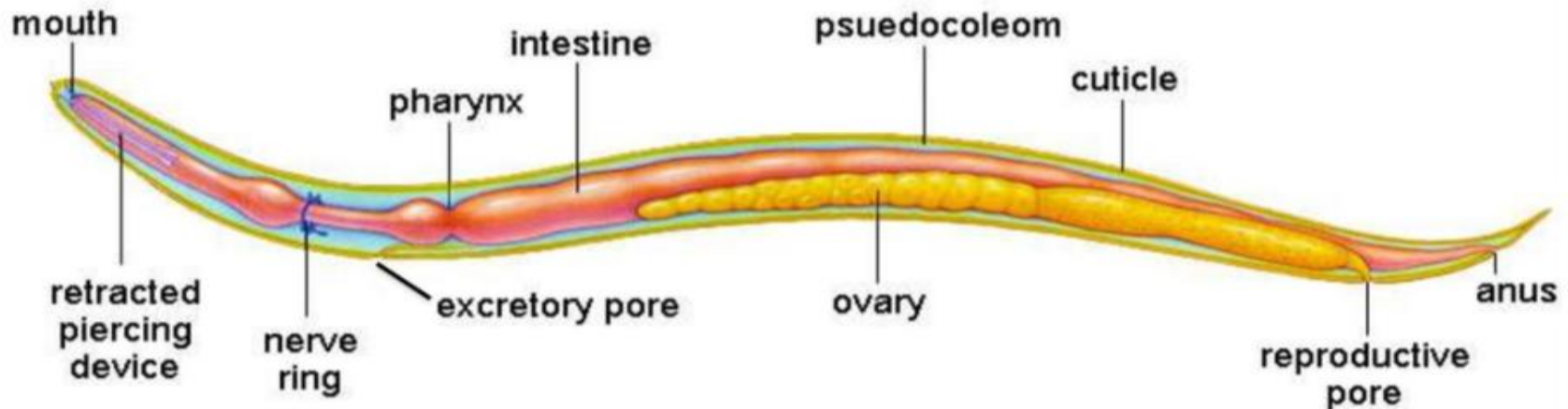


- Marine predators, eat other invertebrates such as worms and crustaceans
- One species (*Lineus longissimus*, 长纵沟纽虫) reaches a length over 50 m making it the longest invertebrate known



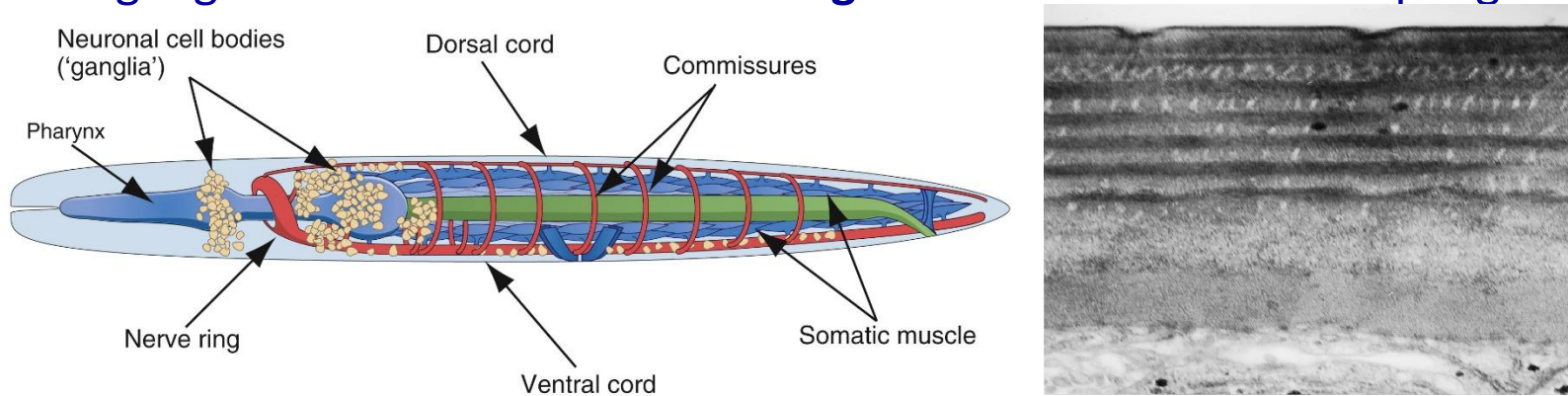
6. Nematodes/Round Worms (线虫, phylum of Nematoda)

- Bilateral symmetry; unsegmented body; consists of three primary germ layers (ectoderm, mesoderm and endoderm)
- **Complete digestive system**, including stomodeum (口道), intestine and proctodeum (原肛道)
 - ❑ The stomodeum consists of the mouth and lips, buccal cavity and the pharynx (esophagus)
 - ❑ The intestine functions by digesting, absorbing water and nutrients, and eliminating the residues of digestion
 - ❑ The proctodeum serves as the anus and is where waste is excreted



● Simple nervous system

- ❑ Consists of a set of neuronal (神经元的) processes that run lengthwise on the nematode body
- ❑ The neuronal processes consist of **neurons** (神经元) that have a cell body also known as a **neurocyte**. A group of neurocytes is called a **ganglion** (神经节)
- ❑ The ganglion connect to the **nerve ring** which surrounds the esophagus



● Have cuticle (角质层/表皮) as the main defense mechanisms

- ❑ The cuticle is the flexible outer covering which acts an **exoskeleton**, allowing the nematode to maintain internal stability
- ❑ It is formed by secretion of the hypodermis (真皮层), the thin cellular layer underling the cuticle
- ❑ The cuticle functions as a barrier, a sensory array, and as a hydrostatic skeleton